

Rasel Meya

AI Engineer | Machine Learning Engineer | Kaggle 2X Expert

- raselmeya2194@gmail.com
- +8801623362194
- Dhaka, Bangladesh
- raselmeya94.github.io/portfolio
- linkedin.com/in/raselmeya
- github.com/raselmeya94



WORK EXPERIENCE

AI Engineer

Tappware Solution Limited

Oct 2023 – Present

Dhaka, Bangladesh

- Conducted R&D in Generative AI, Large Language Models (LLMs), and Natural Language Processing (NLP) to build advanced AI-driven solutions.
- Developed and fine-tuned LLMs for text generation, document summarization, question-answering, and chatbot systems with optimized accuracy and performance.
- Built and deployed NLP models for text classification, sentiment analysis, and information extraction from unstructured data.
- Performed data analysis, feature engineering, and model optimization to ensure scalability and robustness in production environments.
- Collaborated with cross-functional teams to integrate AI/ML solutions across business functions.
- Monitored, evaluated, and continuously improved model performance through fine-tuning and experimentation.
- Designed and implemented AI workflow automation using N8N (both locally with Docker setup and web version), integrating AI models, vector databases, Google Drive, Google Sheets, Email, and WhatsApp APIs.

Machine Learning Engineer Intern

Tappware Solution Limited

Oct 2022 – Jan 2023

Dhaka, Bangladesh

- Conducted R&D on Elasticsearch for ranking and indexing in a Linux environment.
- Documented installation, configuration, and implementation processes of Elasticsearch.
- Contributed to robotic project development and NLP-based Bengali text extraction and indexing.
- Collaborated with team members to troubleshoot and resolve technical challenges.

PROJECTS

Personal Projects

DIALOGUE2SOAP: Medical SOAP Notes Generator (HF | GitHub)

Oct–Nov 2025

Fine-tuned microsoft/Phi-3-mini-4k-instruct model to automatically generate structured SOAP notes from doctor-patient dialogues.

- Applied LLM fine-tuning for medical domain adaptation.
- Automated clinical documentation workflow using prompt engineering and instruction tuning.
- Improved structured medical text generation accuracy and consistency.

Multilingual RAG System – Bengali + English (GitHub)

Jul–Aug 2025

Production-ready Retrieval-Augmented Generation (RAG) pipeline supporting Bengali and English queries using FAISS and MiniLM embeddings.

- Built end-to-end RAG architecture including document ingestion, chunking, embedding generation, and vector search.
- Processed Bengali PDFs with Mistral OCR and custom preprocessing for noise and mixed-language handling.
- Handled challenges like OCR noise, ligature errors, mixed-language content, and complex MCQ structures with custom preprocessing scripts.

GitWithQA – AI-Powered GitHub Q&A Platform (GitHub)

May 2025

LLM-powered system that transforms GitHub repositories into interactive question-answering interfaces for code understanding.

- Implemented repository parsing and embedding generation.
- Built contextual retrieval pipeline for intelligent code Q&A.
- Enabled developers to explore functions, files, and logic through conversational AI interfaces.

Multi-Task LLM-Based Web Application (GitHub | Streamlit)

2025

Unified LLM-powered platform for email generation, text and audio summarization, document conversion, Q&A, and article writing.

- Built multi-task interface supporting diverse NLP workflows in a single Streamlit application.

Lekhoni – Handwritten Bangla Dataset Platform (GitHub) <i>Community-driven platform for collecting handwritten Bangla characters for OCR and AI research.</i> - Built drawing canvas with label selection, real-time stats, and secure backend for dataset collection. - Supports OCR, handwriting recognition, and AI research with scalable data pipeline.	2023
Resume Categorization – NLP & ML (GitHub) <i>Automated resume classification using NLP and ML techniques.</i> - Trained and evaluated classifiers using Random Forest, SVM, ANN, and LSTM. - Developed scripts to process PDFs and categorize resumes efficiently.	2023
Kaggle Machine Learning Projects (Kaggle – Notebook Expert) <i>Diverse ML projects across healthcare, NLP, and computer vision domains.</i> Projects include COVID-19 data analysis, heart failure prediction, autism spectrum disorder detection, breast cancer prediction, mango leaf disease detection, spam classification, Bengali handwritten digit classification, and Bengali multi-label text classification.	2022–2023

Company Projects

Production AI/NLP Systems <i>Backend AI and NLP systems built for production deployment as part of professional work.</i> - AI Summarizer, Invoice Information Extractor, Nothi Application Generator, Certificate Classifier, Content Generator. - YouTube Video Summarizer, Zoom Meeting Summarizer, Resume Analyzer, Bokhshi OCR. - Biennial Transparency Report Analyzer, Voucher Digitizer, Speech-to-Text System, Security Surveillance System. - Built scalable ML pipelines and production-ready APIs across all systems.	Ongoing
AI Resume Prescreening Automation (n8n) (Demo) <i>AI-powered resume screening automation system built with n8n integrating LLM evaluation and workflow automation.</i> - Automated resume ingestion, PDF text extraction, and structured parsing from incoming applications. - Implemented LLM-driven job-role matching and generated a dynamic FitScore for ranking candidates. - Stored structured candidate data in Google Sheets and triggered automated email notifications to HR and applicants.	Dec 2024
Invoice Processing Automation Agent (n8n) (Demo) <i>AI-driven invoice extraction and notification workflow automated using n8n.</i> - Automated invoice retrieval from Google Drive and performed intelligent field extraction using prompt-engineered LLM queries. - Parsed and validated invoice metadata (vendor, amount, date, tax) and stored results in Google Sheets. - Implemented real-time WhatsApp notification system for finance team updates.	Jan 2025
Appointment Booking Automation Agent (n8n) <i>Automated end-to-end appointment booking workflow using n8n with LLM-assisted scheduling logic.</i> Designed conversational booking flow with calendar integration and automated confirmation notifications.	2025
Certification Information Extraction Agent (n8n) <i>Automated extraction of key information from certificates using n8n workflow automation.</i> Integrated OCR and LLM-based parsing to extract fields from TIN, VAT, Trade License, Incorporated, and other certificates.	2025
Certificate Classification and Information Extraction System <i>Web and Android application for real-time image-to-information extraction using machine learning.</i> - Developed Django APIs to classify and extract key information from various certificate types. - Built web and Android app supporting real-time image upload and ML-driven extraction.	2025
X-Road Data Exchange Layer Solution (Article) <i>Enhanced secure data exchange in public sector systems.</i> Implemented best practices for API gateways and interoperability in e-government infrastructure.	

Open Source / PyPI Packages

Bangla Synonyms (PyPI GitHub) <i>Bangla NLP package for programmatic synonym lookup.</i> Developed a Python package providing Bangla synonym retrieval through offline datasets and live web scraping, with source metadata and CLI support.	2025
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pandas2toon (PyPI GitHub)	2024
<i>Data serialization library for pandas and TOON format.</i>	
Built a bidirectional converter between pandas DataFrames and TOON format with support for nested structures, customizable formatting, and flexible I/O.	
Directory Visualizer (PyPI GitHub)	2023
<i>CLI utility for visualizing project directory structures.</i>	
Created a command-line tool that generates Markdown trees and Mermaid diagrams with configurable traversal depth for documenting large codebases.	

Hugging Face Models & Datasets

Hugging Face Model and Dataset Contributions (Hugging Face)	2023–2025
<i>Public models and datasets for Bengali NLP research.</i>	
- Fine-tuned Mistral and DeepSeek models for Bengali multi-label text classification. - Published a Bengali multi-label text dataset for open research, enabling NLP and ML experimentation.	

RESEARCH

Prompt-Based Fine-Tuning of Bangla Language Models for Sequence Classification	2025 – Published
Explored prompt-based fine-tuning for Bengali sentiment analysis and news categorization.	
Achieved 2–4% higher accuracy using only 20% of training data compared to traditional fine-tuning methods.	
DOI: 10.1007/978-981-96-2721-9_27	
Detection of Autism Spectrum Disorder Based on Optimal Feature Selection Using Machine Learning	On Review
Applied feature selection and ML models to predict ASD with 100% accuracy across four age-specific datasets.	
Enhanced early detection and intervention strategies.	
MedicineNet: A Deep Learning Architecture for Recognizing Medicine Names in Handwritten Prescriptions	On Review
Developed CNN and hybrid deep learning models (CNN + Attention + LSTM/BiLSTM/BiGRU).	
- Achieved 91.03% accuracy and 91.85% precision for handwritten prescription recognition. - Improved safety and automation in prescription processing systems.	

EDUCATION

M.Sc. in Computer Science & Telecommunication Engineering	Aug 2021 – May 2023
<i>Noakhali Science and Technology University</i>	
CGPA: 3.62 / 4.0	
<i>Relevant Coursework:</i> Machine Learning, Cloud Computing, Wireless Sensor Network, Stochastic Process and Modeling.	
B.Sc. in Computer Science & Telecommunication Engineering	Jan 2018 – Jan 2021
<i>Noakhali Science and Technology University</i>	
CGPA: 3.29 / 4.0	
<i>Relevant Coursework:</i> Data Structures and C, Object Oriented Programming, Database Management Systems, Software Engineering and Information Systems, Data Mining and Warehousing, Artificial Intelligence and Neural Networking, Image Processing.	

TECHNICAL SKILLS

Programming Languages: Python, R (Basic), C, C++, SQL, JavaScript, HTML, CSS
Frameworks & Tools: LangChain, LangGraph, Django, FastAPI, REST APIs, Streamlit, Gradio
Workflow Automation: N8N (Local Docker Setup & Web), API Integrations, AI Workflow Automation
Databases: Microsoft SQL Server, PostgreSQL, MongoDB
Search Engine: Elasticsearch
ML & AI Libraries: Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, PyTorch, OpenCV, Hugging Face Transformers, sentence-transformers
Generative AI & NLP: Agentic AI, LLMs, RAG, Transformers, NLP, Prompt Engineering, Feature Engineering
Vector Databases: FAISS, ChromaDB
Visualization: Matplotlib, Seaborn, Plotly
Version Control & Tools: Git, GitHub, GitLab